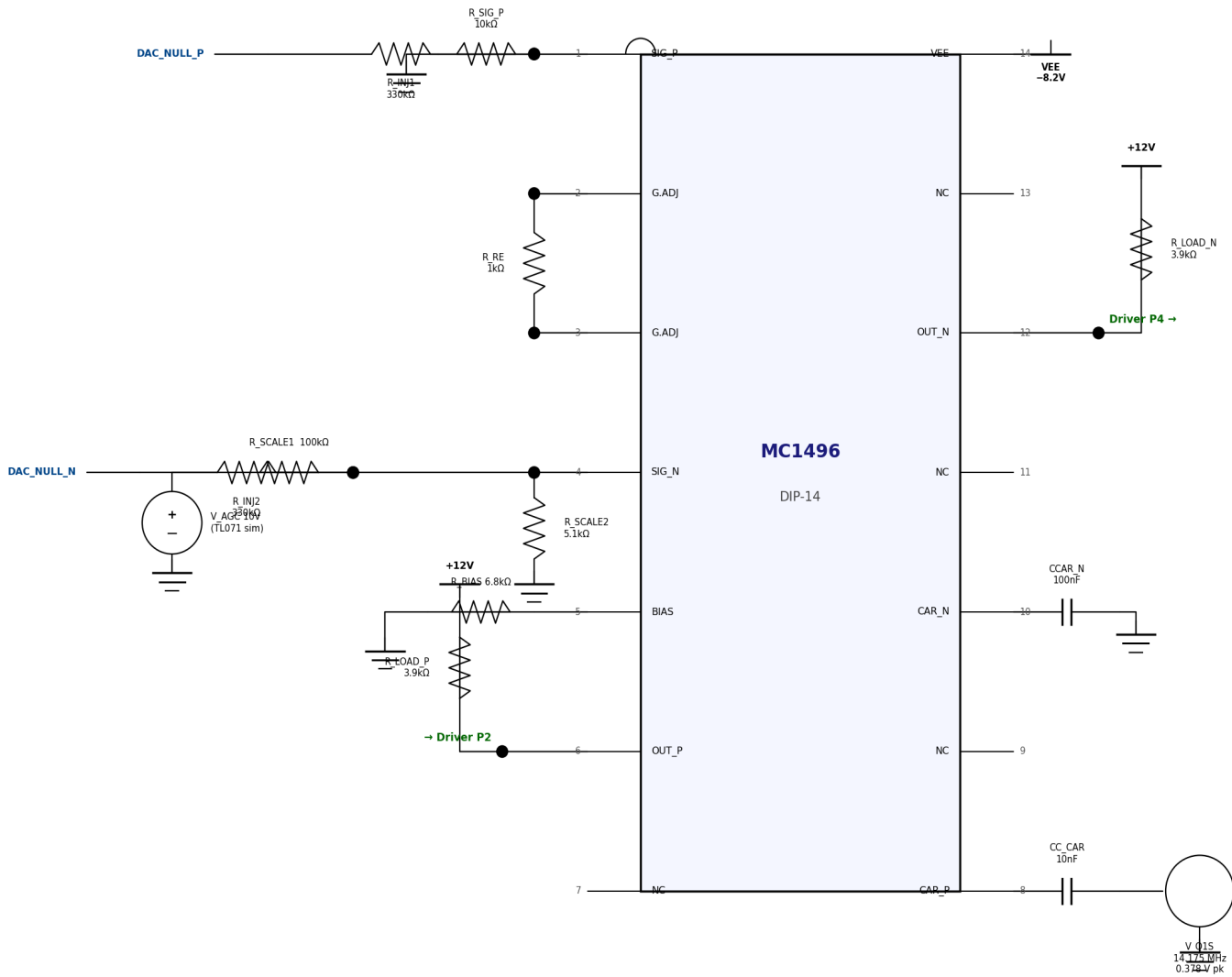


MC1496 VCA Buffer — 20m CW Transmitter

Keyed buffer / amplitude control stage · Pinout per ON Semiconductor MC1496 datasheet



VDD (+12V): enters via R_LOAD_P / R_LOAD_N only

VEE (−8.2V): pin 14 only

GND: R_BIAS (pin 5), signal refs, cap bypasses

V_AGC: sim stand-in for TL071 AGC integrator

Digital null: DAC_NULL_P/N → R_INJ1/2 (330kΩ) → pins 1/4

△ Pin numbers corrected from earlier docs: OUT P = pin 6, OUT N = pin 12 | CAR P = pin 8, CAR N = pin 10 | VEE = pin 14 | No dedicated VCC pin — +12 V through load resistors only | Digital null DAC: R_{INJ1}/R_{INJ2} (330kΩ) replace mechanical null pot

MC1496 VCA Buffer — Schematic Description

20m CW Transmitter · Keyed Buffer / Amplitude Control Stage

For schematic drawing and verification. Pin numbers are per ON Semiconductor MC1496 datasheet.

Supply Rails

Net	Voltage	Notes
VDD	+12 V	Enters only through output load resistors R_LOAD_P / R_LOAD_N — no dedicated VCC pin on MC1496
VEE	−8.2 V	Pin 14 only
GND	0 V	Signal references, bypass caps

MC1496 DIP-14 Pin Assignments

Source: ON Semiconductor MC1496 datasheet. Earlier design documents had pins 6/8/10/12 swapped — this table is correct.

Pin	Datasheet Function	Circuit Net / Connection
1	Signal input (+)	SIG_P → R_SIG_P (10 kΩ to GND); also receives R_INJ1 (330 kΩ) from DAC_NULL_P
2	Gain adjust	Pin 2 end of R_RE (1 kΩ between pins 2 and 3)
3	Gain adjust	Pin 3 end of R_RE (1 kΩ between pins 2 and 3)
4	Signal input (−)	SIG_N → AGC divider (R_SCALE1 / R_SCALE2); also receives R_INJ2 (330 kΩ) from DAC_NULL_N
5	Bias	VBIAS → R_BIAS (1 kΩ) to VEE
6	Output (+)	OUT_P → R_LOAD_P bottom (3.9 kΩ to VDD) → driver P2
7	NC	—
8	Carrier input	CAR_P → CC_CAR (10 nF) → VFO RF input (14.175 MHz)
9	NC	—
10	Carrier input	CAR_N → CCAR_N (100 nF to GND)
11	NC	—
12	Output (−)	OUT_N → R_LOAD_N bottom (3.9 kΩ to VDD) → driver P4
13	NC	—
14	Vee	VEE rail (−8.2 V)

Blue-bold = output pins. Green-bold = carrier input pins.

All Components and Connections

RF Carrier Input Path

Ref	Value	Connection
V_Q1S	14.175 MHz, 0.378 V pk	Source (+) → _rf_in node; (-) → GND
CC_CAR	10 nF	_rf_in → CC_CAR → CAR_P (pin 8)
CCAR_N	100 nF	GND → CCAR_N → CAR_N (pin 10) [bypass, keep short]

Output Load Resistors (VDD to MC1496 outputs)

VDD must connect only to R_LOAD_P and R_LOAD_N tops. It shares NO node with the RF carrier input path.

Ref	Value	Connection
R_LOAD_P	3.9 kΩ	Top → VDD (+12 V); Bottom → OUT_P (pin 6) → driver P2
R_LOAD_N	3.9 kΩ	Top → VDD (+12 V); Bottom → OUT_N (pin 12) → driver P4

Signal Port / AGC Network

Ref	Value	Connection
R_SIG_P	10 kΩ	SIG_P (pin 1) → GND [holds pin 1 reference near GND]
R_SCALE1	100 kΩ	V_AGC (+) → R_SCALE1 → SIG_N (pin 4) [AGC upper divider leg]
R_SCALE2	5.1 kΩ	SIG_N (pin 4) → GND [AGC lower divider leg; ratio 4.85%]
V_AGC	10 V DC	(Simulation stand-in for TL071 AGC integrator output) (+) → R_SCALE1 bottom; (-) → GND

Digital Carrier Null Network

Replaces mechanical trimmer pot. Two DAC outputs inject a differential offset to cancel residual carrier caused by transistor mismatch inside the MC1496. Target: ≥65 dB suppression. DAC_NULL_P and DAC_NULL_N are complementary 12-bit outputs; midscale = no injection; firmware binary-searches for minimum carrier at key-down.

Ref	Value	Connection
R_INJ1	330 kΩ	DAC_NULL_P → R_INJ1 → SIG_P (pin 1); joins existing R_SIG_P / pin 1 junction
R_INJ2	330 kΩ	DAC_NULL_N → R_INJ2 → SIG_N (pin 4); joins existing AGC / pin 4 junction

Full-scale swing: ±16 mV per pin, ±33 mV differential total — covers MC1496 worst-case input offset (≤5 mV). Step size at 12-bit: ~8 μV. No series protection resistors needed; 330 kΩ inherently limits injection current. Null stored in NVRAM, refreshed periodically.

Bias and Gain-Set

Ref	Value	Connection
R_BIAS	6.8 kΩ	VBIAS (pin 5) → R_BIAS → GND [sets I5 ≈ 1 mA per AN531/D formula]
R_RE	1 kΩ	Between pin 2 and pin 3 (NOT shorted to VEE); sets gain $A_v = R_{LOAD} / R_{RE}$;

Signal Flow Diagram (ASCII)

[illegible]

Key Layout / Drawing Notes

- **Pin 6 = Output, Pin 8 = Carrier input.** Earlier documents had these swapped. Verify against datasheet before drawing.
- VDD connects **only** to R_LOAD_P top and R_LOAD_N top. It shares no node with the RF carrier input (CC_CAR / pin 8).
- Pin 6 (OUT_P) and pin 8 (CAR_P) are on the same side of the DIP but are **completely separate nets**.
- Pins 2 and 3 are both gain-adjust — connect R_RE (1 kΩ) between pin 2 and pin 3. Do NOT short to VEE.
- No VCC pin on the MC1496. +12 V enters only through R_LOAD_P and R_LOAD_N.
- Vee = pin 14, not pin 7. Pin 7 is NC.

- CCAR_N (100 nF) bypasses pin 10 to GND — keep lead very short.
- **Digital carrier null:** R_INJ1 (330 k Ω) from DAC_NULL_P to pin 1; R_INJ2 (330 k Ω) from DAC_NULL_N to pin 4. Complementary 12-bit DAC codes; midscale = no injection; firmware binary-searches for null at key-down. Null stored in NVRAM. No separate protection resistors needed — 330 k Ω limits current inherently.
- OUT_P and OUT_N feed the driver push-pull grids directly (no interstage transformer).